Adaptive Governance in Tipping Markets: An Agent-Based Model of Policy and Network Topology in U.S. EV Charging Standardization

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Abstract

The transition from the Combined Charging System (CCS1) to the North American Charging Standard (NACS) presents a live test case of Arthur's (1989) theory of path dependence and increasing returns. This paper develops an Agent-Based Model (ABM) to examine how government subsidy design and physical network topology jointly shape the outcome of this standards competition. A 2×2 experimental matrix (Static vs. Adaptive policy; Linear Grid vs. Scale-Free Hub topology) was tested across 200 simulation runs grounded in empirical U.S. charging infrastructure data and calibrated to real policy milestones. Results show that Adaptive policy significantly reduces stranded-asset waste relative to Static policy, but this advantage is entirely conditional on topology: it emerges clearly in Scale-Free Hub networks but disappears in Linear Grid networks, where the market never approaches the threshold needed to trigger a policy pivot. Notably, Adaptive policy did not reduce — and was numerically associated with a modest increase in — short-term consumer friction, though this difference was not statistically significant, suggestive of a transitional cost of accelerated standard-switching for legacy users. Furthermore, rather than universally accelerating diffusion, Scale-Free Hub topologies acted as structural catalysts that substantially widened the variance of possible outcomes, illustrating the non-ergodic dynamics Arthur describes. These findings suggest that effective governance of tipping markets depends on the interaction between policy design and spatial infrastructure structure, not on subsidy mechanisms alone.

Keywords: path dependence; increasing returns; standards competition; agent-based modeling; stranded assets; network topology; electric vehicle infrastructure

JEL Classification: L51, L98, Q48, D85

Introduction

Technology standards competitions do not always resolve toward the objectively superior option. Arthur (1989) argues that when technologies exhibit increasing returns to adoption, small early advantages can compound into market lock-in—a dynamic now visible in the U.S. transition from CCS1 to NACS. This transition is particularly consequential because transportation generates about 29% of U.S. greenhouse gas emissions, while battery electric vehicles (BEVs) produce an estimated 54% fewer lifetime emissions than internal combustion vehicles and 32% fewer than plug-in hybrids (Climate Adaptation Platform, 2026; Esmaili et al., 2024; U.S. EPA, 2023). Yet despite this urgency, only 1.9% of the 291 million registered U.S. vehicles are electric (Montoya & O'Dell, 2026).

Despite its environmental benefits, EV diffusion is severely constrained by coordination failures between infrastructure deployment and vehicle adoption. Scholars identify standardization as the primary solution to this market failure, as uniform standards ensure interoperability, prevent infrastructure "balkanization," and build consumer confidence (Muto, 2017). To combat this, the U.S. government has intervened with substantial federal funding programs, such as the National Electric Vehicle Infrastructure (NEVI) Formula Program and the Charging and Fueling Infrastructure (CFI) grant program (Joint Office of Energy and Transportation, 2026).

This transition sits atop a high-stakes technological competition over charging standards, particularly between the Combined Charging System (CCS1) and the North American Charging Standard (NACS). The charging connector market is evolving rapidly; while much of the early 2020s was dominated by CCS1/2, since 2023, the industry has shifted decisively towards NACS. Starting with Ford and General Motors, the vast majority of major automakers—including Hyundai Motor Group, Toyota, Volkswagen Group, Stellantis, and BMW Group—have introduced their transition to NACS-equipped vehicles starting in the 2025 model year, with existing CCS vehicles gaining access to Tesla's charging network through adapters (Beckford, 2025).

This rapid market transition creates a central policy challenge. While government intervention is needed to overcome early market inertia, static policies risk generating economic waste if they become misaligned with consumer preferences. A major concern is the creation of stranded assets, imposing substantial costs on governments, firms, and consumers (Muto, 2017). This paper argues that interventions in tipping markets must be adaptive rather than static. Testing Arthur's (1989) theories of path dependence and technological lock-in against simulated regulatory frameworks, it examines whether adaptive governance can reduce stranded investments and support a more efficient transition to electric mobility.

Background and Motivation

Arthur in 1989 introduced the idea of lock-in occurring through a radically different approach, increasing returns to adoption. He explains; when two or more technologies compete in a market, initial historical events can give one technology an early, arbitrary lead. This lead results in a positive feedback loop as the leading technology gains a larger installed base. Hence, improving efficiency, reducing production costs through economies of scale, and naturally appealing to a broader proportion of future adopters, ultimately driving the market toward a monopolistic 'lock-in' (Arthur, 1989).

The radical implication of Arthur's framework is the concept of path dependence. In a free market with little to no intervention, markets can become locked into a suboptimal path (QWERTY keyboard) because of "excess inertia" and the coordination costs required to switch systems entirely (Farrell & Saloner, 1985; David, 1985). The recent shift toward the North American Charging Standard (NACS) as SAE J3400 presents an interesting contrast to the QWERTY paradox, in that the market appears to be coalescing around a technically favoured product. Compared to the Combined Charging System (CCS1), NACS is smaller, lighter, and reuses AC pins for DC charging via a single-wire CAN specification, avoiding some of the bulkier hardware associated with CCS1 (Paryani, 2023).

Beyond physical ergonomics, the transition to NACS is also associated with a documented disparity in network reliability. The Tesla Supercharger network, which utilizes NACS, reports an operational uptime of more than 99%, with 4% of users reporting major difficulties, while CCS networks report that over 25% of CCS chargers in key regions were entirely non-functional (Paryani, 2023). This reliability gap plausibly drives the observed shift in automaker and consumer preference toward NACS, though whether it alone can overcome CCS1's installed base within a realistic policy timeframe remains an empirical question addressed by this paper's simulations.

The battleground in this context is amplified by the highly interdependent arena of a two-sided market and constrained by a "chicken-and-egg" dilemma. Consumer demand for EVs is heavily dictated by the availability of a robust public charging network, while private investment is dependent upon a circulating base of EVs to ensure station profitability (Springel, 2021). The magnitude of this indirect network effect appears substantial: empirical modeling by Li et al. (2017) estimates that a 10% increase in public charging stations is associated with an 8% increase in EV sales. Using Norwegian municipal panel data, Van Dijk et al. (2022) similarly find that the elasticity of EV adoption with respect to charging infrastructure rises as the pre-existing network grows, an early empirical signature of the same increasing-returns dynamic this paper's ABM is built to simulate. Consequently, Springel (2021) estimates that subsidizing charging station deployment is more than twice as cost-effective in promoting EV adoption as injecting equivalent funds directly into consumer vehicle subsidies.

To capture this multiplier effect, the U.S. federal government authorized $5 billion USD through NEVI and $2.5 billion USD through the CFI grant program, with NEVI recommending stations be spaced no more than 50 miles apart along designated highway corridors (Abramson & McFarlane, 2023). However, NEVI's requirement that every federally funded DC Fast Charger include at least four CCS1-equipped ports ties the program to a legacy standard regardless of market shifts.

Mandating the deployment of CCS1 stations while major automakers pivot to NACS creates a meaningful risk of "stranded assets" (Muto, 2017), particularly given that equipment and installation costs for a single four-port DC Fast Charging station range from $720,000 to $1.2

million USD (Abramson & McFarlane, 2023). In an environment where proposed legislation threatens to reduce federal charging infrastructure deployment (Electrification Coalition, 2026), a static mandate that becomes misaligned with market-preferred standards risks generating substantial economic waste.

While Arthur's (1989) theory explains why a market locks in, network topology may help explain how quickly that lock-in occurs. EV charging infrastructure does not diffuse evenly across a blank geographical canvas; rather, it tends to proliferate through clustered spatial agglomerations (Esmaili et al., 2024; Carlton & Sultana, 2022).

Because charging infrastructure relies on "scale-free hubs" (highly clustered, dominant central nodes characterized by a power-law degree distribution where a small minority of sites possess a massive number of connections (Huang et al., 2022)) in high-traffic urban areas for profitability, utilization varies substantially: hubs in Las Vegas and Miami average 43% utilization during peak daytime hours (Paren, 2025), while rural stations often struggle to break even. The U.S. Fast Charger dataset reflects this pattern, including 96-port Tesla NACS hubs in California.

In network-diffusion theory, high-degree nodes in a scale-free network can act as accelerants, exposing a large volume of consumers to a given technology simultaneously (Huang et al., 2022). This suggests one plausible hypothesis: that concentrating charging deployment in scale-free hubs could compress the timeframe for standard diffusion relative to a more evenly distributed linear grid (a uniform spatial distribution characterized by strictly localized, point-to-point connectivity, long average path lengths, and a complete absence of centralized high-density nodes (Carlton & Sultana, 2022)) . Whether this structural advantage is realized in practice, and whether it raises typical outcomes or widens their range, is examined in Hypothesis 3.

This study of the EV charging standards competition extends into the broader domain of complexity economics, which, as W. Brian Arthur's *Complexity and the Economy* outlines, frames economies as complex, adaptive, and evolving systems (Arthur, 1999). Traditional closed-form equilibrium models are methodologically inadequate for this context, as they mathematically cannot capture out-of-equilibrium, path-dependent, and nonlinear dynamics (Farmer & Foley, 2009). Because EV infrastructure diffusion is driven by increasing returns, it constitutes a

non-ergodic market process that requires an agent-based computational economics approach to observe how macroeconomic regularities emerge strictly from local micro-interactions without assumed equilibrium states (Tesfatsion, 2006). Consequently, utilizing an Agent-Based Model (ABM) constructed in NetLogo is methodologically essential, as it computationally simulates the exact out-of-equilibrium transitional forces and spatial tipping dynamics governing this market.

Hypothesis

Building upon the theoretical frameworks of path dependence and network externalities, this research utilizes an Agent-Based Model (ABM) to test a 2x2 experimental matrix. The simulation evaluates two regulatory approaches (Static vs. Adaptive Policy Modes) against two distinct spatial environments (Scale-Free Hubs vs. Linear Grid Topologies). The expectations for this simulation are defined by the following three hypotheses:

Hypothesis 1:

Under a Static policy framework that rigidly subsidizes CCS1 infrastructure, cumulative subsidy waste is expected to rise as the market moves toward the North American Charging Standard (NACS), driven by the increasing returns to adoption described by Arthur (1989). Because this misalignment between public investment and consumer preference should be more pronounced wherever network structure allows the market to move furthest, the magnitude of stranded-asset accumulation under Static policy is expected to vary by network topology, with an accompanying rise in consumer friction (`total-trips-failed`).

Hypothesis 2:

Conversely, an Adaptive policy mode that dynamically tracks national market share and pivots subsidies to NACS upon crossing a critical adoption threshold is expected to reduce capital stranding relative to Static policy — but only in conditions where the underlying market moves enough to cross that threshold. Where topology constrains diffusion below the threshold, Adaptive and Static policies are expected to perform similarly, since the pivot mechanism will not have an

opportunity to activate. Where the pivot does activate, total-trips-failed is not assumed to fall automatically, since redirecting subsidies away from CCS1 may itself impose short-term costs on consumers still using that standard.

Hypothesis 3:

Regarding spatial diffusion, it is hypothesized that Scale-Free Hub topology, by virtue of its high-degree "super-spreader" nodes (Huang et al., 2022), will alter the distribution of diffusion outcomes relative to a Linear Grid — plausibly by raising the typical (central-tendency) outcome, by widening the range of possible outcomes, or both. A central prediction of increasing-returns theory (Arthur, 1989) is that such systems are non-ergodic, so this hypothesis anticipates that topology's effect may show up more clearly in the variance and ceiling of outcomes than in their median.

Data

To ensure the Agent-Based Model (ABM) captures the microfoundational dynamics of technology diffusion under policy and topological variation, the NetLogo simulation is grounded in empirical data from the U.S. Department of Energy's Alternative Fuels Data Center (AFDC). The timeline is calibrated to real-world policy milestones from the AFDC's Federal and State Laws database, with each tick representing one week. The simulation begins at Tick 0 (November 2021), corresponding to the Bipartisan Infrastructure Law and the launch of the National Electric Vehicle Infrastructure (NEVI) Formula Program (Abramson & McFarlane, 2023). A milestone is hardcoded at Tick 104 (December 2023), marking federal recognition of the North American Charging Standard (NACS) as SAE J3400. This milestone allows the "Adaptive" policy mode to shift public funding from CCS1 when national market share reaches the model's 25% adaptive threshold, calibrated to be achievable within the twenty-year observation window.

To capture the market scarcity and "range anxiety" driving evolutionary technological pressure, the ABM uses representative scaling. The simulation imports a randomized sample of 1,200

fast-charging stations that preserves the national density distribution and normalizes the consumer population to 1,500 circulating EV agents.

The two spatial topologies are generated algorithmically rather than imposed geometrically. In the Scale-Free Hubs condition, each station preferentially links to nearby, highly connected stations within a fixed radius, producing hub-and-spoke clustering. In the comparison condition, each station links only to its nearest unconnected neighbor, forming a nearest-neighbor chain (hereafter "Linear Grid" for consistency with the hypotheses). Disconnected clusters are then bridged to the giant component, ensuring full network connectivity for the station-conversion mechanism.

Operationalizing bounded rationality, consumer agents lack global omniscience; their search radius is strictly limited to 3 patches, forcing them to base charging decisions exclusively on their local network neighborhoods (Wolbertus et al., 2021). This spatial constraint is coupled with a battery drain rate of 10 units per tick and a maximum charge of 100, so agents require charging about every ten weeks. This makes an incompatible plug a genuine stranding event, triggering the evolutionary replacement cycle at a plausible cadence.

In calibrating the diffusion mechanics, this model deliberately departs from unweighted node-counting, which generates artificial "excess inertia" (Farrell & Saloner, 1985). To operationalize structural network gravity, the model extracts the specific port capacity (`EV DC Fast Count`) for each station from the empirical dataset. Drawing on Arthur's (1989) framework of heterogeneous technology payoffs, the model evaluates the Effective Reliable Capacity of the local network: NACS port capacity is weighted at a 1.00 multiplier to reflect its superior uptime, while CCS1 capacity is discounted to 0.75 to account for its empirically observed non-functional rate (Paryani, 2023). This weighting is applied symmetrically to both the demand-side consumer preference mechanism and the supply-side station-conversion mechanism, and is intended to allow high-density "Scale-Free Hub" topologies to exert greater structural influence over consumer preference than an unweighted count would permit, rather than to guarantee any particular market outcome.

The model additionally distinguishes between two levels of economic rationality operating simultaneously, following the local-versus-global logic of network diffusion theory. At the micro

level, individual charging stations facing sustained unprofitability convert toward whichever standard commands greater weighted capacity within their immediate network neighborhood (not the national market as a whole) allowing localized clusters to tip toward NACS independently of, and in advance of, the aggregate national share. A station must register as unprofitable for two consecutive annual cycles before conversion is triggered, introducing a deliberate frictional delay against instantaneous supplier reaction. At the macro level, by contrast, the federal government's Adaptive policy pivot is governed exclusively by the global national market-share threshold described above, reflecting the comparatively slower, aggregate-data-driven nature of federal policymaking relative to individual firm behavior.

On the supply side, the government-subsidy parameter is set to 0.10, covering part of a subsidized station's operating shortfall. This was calibrated below the fixed maintenance cost of 0.15 so that subsidized stations can still become unprofitable; an earlier value of 0.80, matching NEVI's 80% cost-share (Abramson & McFarlane, 2023), made subsidized stations effectively immune to conversion, preventing a meaningful test of Hypotheses 1 and 2. To measure economic destruction under static policies, the model applies a 5% minimum viable utilization threshold over each 52-tick business cycle, with subsidies to stations below this threshold recorded as wasted. A fixed retrofit cost of 50 is also added whenever a subsidized station converts to the rival standard, representing stranded assets (Muto, 2017).

Model and Results

4.1 Hypotheses 1 & 2: Policy Effects on Stranded Assets

Across the full 200-run experiment, Adaptive policy produced significantly lower cumulative subsidy waste than Static policy ($M = 9{,}655$ vs. $M = 14{,}331$; Welch's $t(173.8) = -4.44$, $p = 1.58 \times 10^{-5}$).

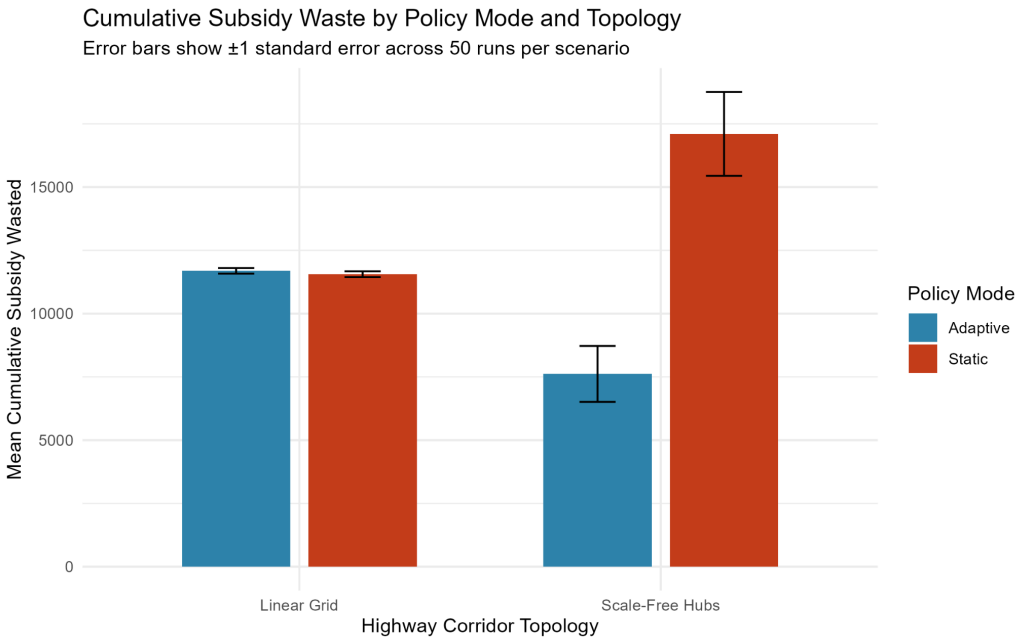


FIGURE 1 — Graph Subsidy Waste

Decomposed by topology (Table 1), results differ sharply between the two conditions. Within Scale-Free Hubs, Adaptive policy reduced waste by more than half relative to Static (M = 7,618 vs. M = 17,103; t(85.4) = -4.76, p = 7.79×10^{-6}, d ≈ 1.03). Within the Linear Grid, no significant difference emerged (M = 11,692 vs. M = 11,559; t(97.9) = 0.84, p = 0.405, d ≈ 0.17) (Table 1). Across all 100 Linear Grid runs, NACS market share never exceeded approximately 20% (Section 4.2), below the 25% adaptive-threshold required for the policy pivot to activate.

Comparison	M (Adaptive)	M (Static)	t	df	p
Overall	9,655	14,331	-4.44	173.8	1.58e-05
Linear Grid	11,692	11,559	0.84	97.9	0.405
Scale-Free Hubs	7,618	17,103	-4.76	85.4	7.79e-06

TABLE 1 — Subsidy Waste t-test Results by Topology

The temporal trajectory of waste accumulation (Figure 2) mirrors this pattern. In Linear Grid, Adaptive and Static trajectories track one another closely across the full 1,040-tick window. In Scale-Free Hubs, the trajectories diverge sharply beginning around tick 150, coinciding with the earliest station-level conversions permitted under the model's two-year unprofitability threshold.

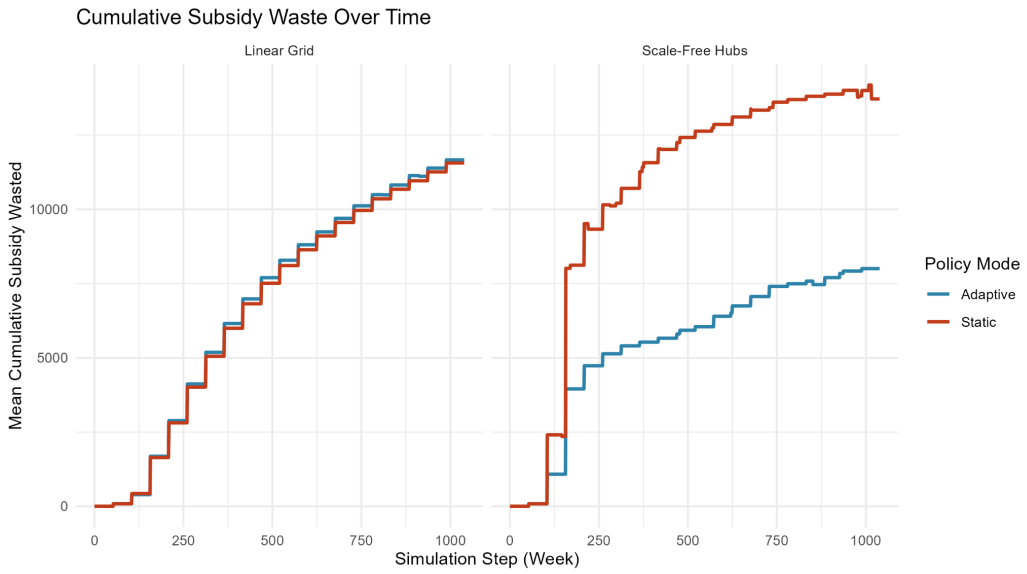


FIGURE 2 — Waste Trajectory

Consumer friction, operationalized as `total_trips_failed`, did not follow the hypothesized pattern. No significant difference emerged between policies (M = 1,782 under Adaptive vs. M = 1,190 under Static; t(137.8) = 1.25, p = 0.214), and the direction was opposite to that hypothesized, with Adaptive numerically higher (Table 2, Figure 3).

Comparison	M (Adaptive)	M (Static)	t	df	p
Total Trips Failed	1,782	1,190	1.25	137.8	0.214

TABLE 2 — Consumer Friction t-test Result

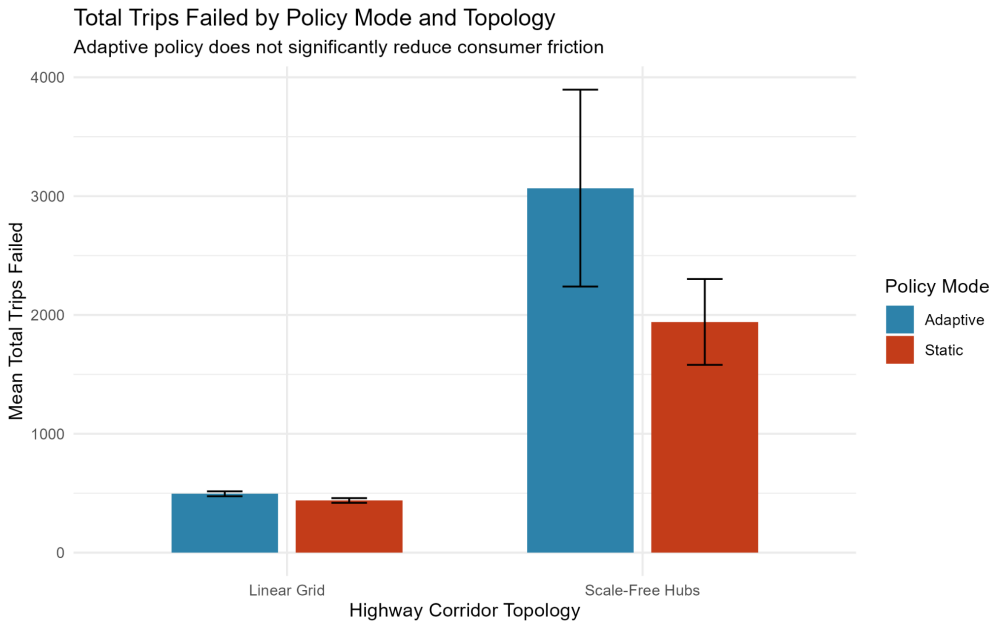


FIGURE 3 — Consumer Friction

4.2 Hypothesis 3: Topological Effects on Diffusion

Median final NACS market share was slightly *lower* under Scale-Free Hubs than under Linear Grid in both policy modes (Wilcoxon rank-sum tests: Static, $W = 1{,}686$, $p = 0.00268$; Adaptive, $W = 1{,}844$, $p = 4.28 \times 10^{-5}$).

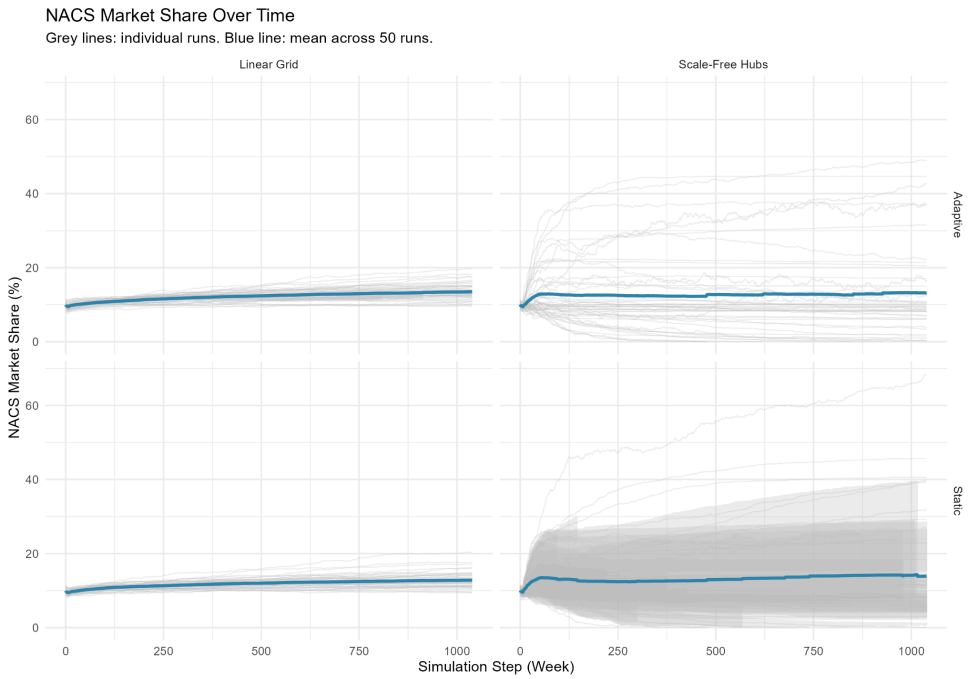


FIGURE 4 — SCurve

Scale-Free Hubs exhibited dramatically greater variance in final outcomes than Linear Grid (F-tests: Static, $F = 0.023$, $p < 2.2 \times 10^{-16}$; Adaptive, $F = 0.034$, $p < 2.2 \times 10^{-16}$), with Linear Grid's variance amounting to only 2–3% of Scale-Free Hubs' variance under both policies (Table 3, Figure 5). The two topologies also differed sharply in ceiling outcomes: Linear Grid's maximum observed NACS share across 100 runs never exceeded 20.3%, while Scale-Free Hubs reached as high as 68.5%.

Policy	Wilcoxon W	Wilcoxon p	Variance ratio (LG:SFH)	F-test p
Static	1,686	0.00268	0.023	<2.2e-16
Adaptive	1,844	4.28e-05	0.034	<2.2e-16

TABLE 3 — Topology Variance and Median Comparison

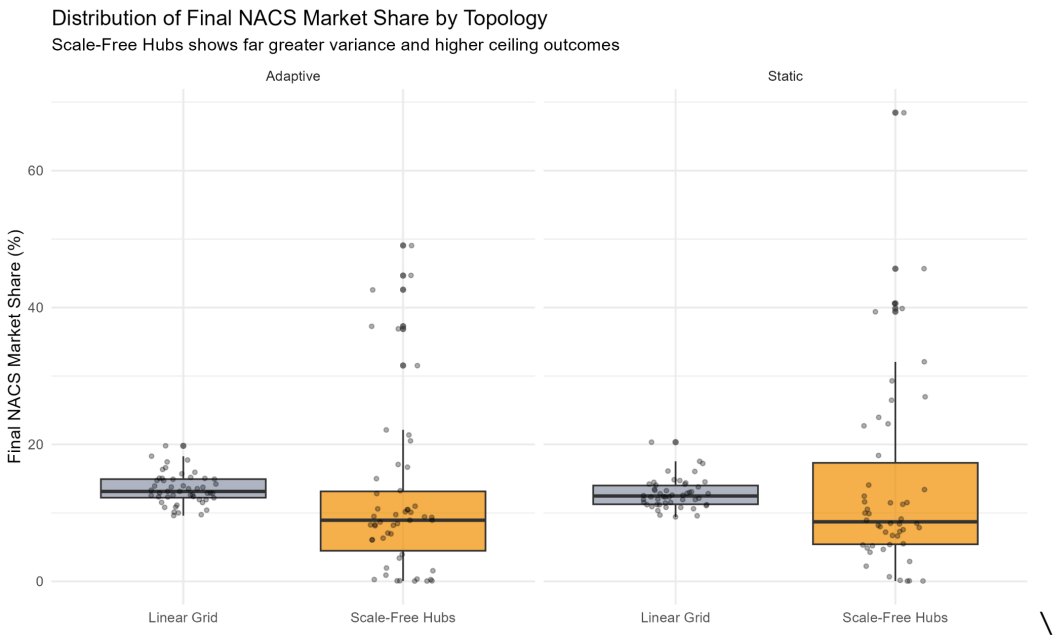


FIGURE 5 — Variance Spread

Test	Raw p	Holm-adjusted p	Significant (α = .05)
Waste, Overall	1.58e-05	7.90e-05	Yes
Waste, Linear Grid	0.405	0.428	No
Waste, Scale-Free Hubs	7.79e-06	4.67e-05	Yes
Consumer Friction	0.214	0.428	No
H3 Wilcoxon, Static	0.00268	0.00804	Yes
H3 Wilcoxon, Adaptive	4.28e-05	1.71e-04	Yes
H3 F-test, Static	2.2e-16	1.76e-15	Yes
H3 F-test, Adaptive	2.2e-16	1.76e-15	Yes

TABLE 4 — Holm-Bonferroni Correction Across All Reported Tests

All statistically significant results reported above remain significant after applying a Holm-Bonferroni correction across the eight hypothesis tests conducted in this study (Table 4). A final cross-cutting observation qualifies both results above: complete market lock-in was rare under

the conditions tested. Only 1 of 200 runs (Scale-Free Hubs, Static policy) crossed a 50% NACS market-share threshold, reaching 68.5%; no run reached full (100%) lock-in within the twenty-year observation window.

4.3 Sensitivity Analysis

To assess whether the central findings depend on parameters calibrated through iterative testing rather than external data (the adaptive-threshold and reliability-weight values noted in Limitations), a supplementary sensitivity analysis was conducted within Scale-Free Hubs, where the Adaptive policy effect was concentrated (20 repetitions per condition).

Varying the adaptive-threshold from 15% to 35% under Adaptive policy produced cumulative waste values of 6,114 (15%), 8,541 (20%), 10,531 (25%, the baseline), 10,798 (30%), and 8,773 (35%). Waste rose from 6,114 at 15% to a peak of 10,798 at 30%, before falling to 8,773 at the highest tested value (35%); standard errors for each threshold are reported in Table 5. The baseline value of 25% used throughout the main analysis falls within this tested range.

Threshold	Mean Waste	SE	Mean NACS Share	n
15%	6,114	1,707	14.9%	20
20%	8,541	1,858	9.8%	20
25% (baseline)	10,531	1,841	13.6%	20
30%	10,798	2,814	8.7%	20
35%	8,773	1,770	5.7%	20

TABLE 5 — Adaptive Policy Waste Across Adaptive-Threshold Values

Varying the CCS1 reliability weight (0.65, 0.75 baseline, 0.85) while comparing both policy modes shows that Adaptive policy produced lower cumulative waste than Static policy at every tested value (Table 6). The Adaptive-Static gap was smaller at 0.65 (4,948 vs. 8,950) than at 0.75 (4,052 vs. 14,827) or 0.85 (4,221 vs. 15,165).

CCS1 Weight	Adaptive	Static	n (each)
0.65	4,948	8,950	20
0.75 (baseline)	4,052	14,827	20
0.85	4,221	15,165	20

TABLE 6 — Adaptive vs. Static Waste Across CCS1 Reliability Weights

Discussion

The hypotheses presented above reflect a conditional formulation developed after initial model runs revealed the topology-dependence of the Adaptive policy pivot; the original hypotheses and the reasoning behind their revision are disclosed in Appendix A for transparency.

The Conditional Success of Adaptive Governance (Hypotheses 1 & 2)

The model provided partial support for the financial expectations of Hypotheses 1 and 2. Overall, Adaptive policy significantly reduced stranded-asset accumulation relative to Static (M = 9,655 vs. M = 14,331, $d \approx 0.67$), but this effect was entirely conditional on spatial topology: within the Linear Grid, there was no statistically detectable difference between policy modes.

This asymmetry has a mechanical rather than substantive explanation. The Adaptive policy pivots subsidies only once NACS crosses a 25% global market-share threshold; in the Linear Grid runs, NACS share never exceeded approximately 20% across 100 simulations, so the triggering condition was simply never met. This suggests that the effectiveness of macro-level adaptive policy is contingent on micro-level spatial structure: a government can design a highly responsive, trigger-based mechanism, but if the underlying physical network never allows the technology to approach critical mass, that mechanism cannot activate, and stranded public capital (Muto, 2017) accumulates much as it would under a static regime.

The Transitional Cost of Standard-Switching (Hypotheses 1 & 2)

The second component of Hypotheses 1 and 2 predicted that Adaptive policy would reduce consumer-facing friction. The results did not support this: `total_trips_failed` was not significantly different between policies, and was numerically higher under Adaptive (1,782 vs. 1,190). No run reached full market lock-in within the twenty-year window.

This plausibly reflects a genuine transitional cost of resolving a standards war rather than a model error. Farrell and Saloner (1985) note that an industry "once firmly bound together by the benefits of compatibility or standardization[,] will be inclined to move extremely reluctantly to a new and better standard because of the coordination problems involved" (p. 71). When the Adaptive pivot activates here, public capital redirects from CCS1 to NACS while a substantial population of CCS1-preferring consumers remains active; as subsidized stations convert, infrastructure previously compatible with these consumers becomes unavailable, generating new stranding events even as aggregate financial waste declines. This points to a real policy tradeoff: accelerating a standard-switch, even one that reduces long-run financial waste, may impose short-term costs on the legacy user base that a purely financial metric does not capture.

Non-Ergodicity and Network Topology (Hypothesis 3)

The evaluation of Hypothesis 3 initially appears to contradict expectations: median final NACS share was slightly lower under Scale-Free Hubs than under the Linear Grid. However, central-tendency statistics are a poor lens for increasing-returns systems, which Arthur (1989) describes as fundamentally non-ergodic: "[c]ompetitions between technologies may have multiple potential outcomes," and "the small events that determine [adoption timing] decide the path of market shares; the process is non-ergodic or path-dependent" (pp. 116, 122). The more informative result is variance. Linear Grid produced a narrow distribution, with no run exceeding roughly 20% NACS share, whereas Scale-Free Hubs exhibited much greater variance, reaching outcomes as high as 68.5%. Consistent with the model's local conversion logic, early random events at hub-adjacent stations can cascade: where they favor CCS1, hubs reinforce the incumbent; where they favor NACS, hubs amplify its spread. As Arthur notes, "[i]nsignificant circumstances become magnified by positive feedback to 'tip' the system into the actual outcome 'selected'" (p. 127). This divergence

is also consistent with the complexity-economics view of markets as emergent, self-organizing systems rather than equilibrium-seeking ones (Arthur, 1999; Farmer & Foley, 2009; Tesfatsion, 2006).

At the same time, these findings contrast with the Background's expectation of rapid NACS adoption. Even under Scale-Free Hubs, the median remained close to the initial seed value, and full lock-in never occurred across 200 runs. A technical advantage alone may therefore be insufficient to overcome a large incumbent installed base within a realistic policy timeframe.

Robustness to Calibrated Parameters (Section 4.3)

Across the tested reliability-weight range (0.65 - 0.85), Adaptive policy produced lower waste than Static at every value, confirming the finding does not depend on the specific 0.75 discount for CCS1's reliability, though the advantage narrows as the discount relaxes toward 0.65. The adaptive-threshold sweep is less conclusive: waste rose from 15% to 30% before falling at 35%, consistent with a higher threshold delaying the pivot and allowing more waste to accumulate before it activates. Given the wide standard errors at this repetition count (n = 20), this decline should be read as inconclusive rather than a reversal of the relationship.

Limitations

Several limitations bear on the interpretation of these results. First, the reliability weights (1.00 for NACS, 0.75 for CCS1) and the adaptive-threshold parameter (25%) were calibrated through iterative testing rather than derived directly from external data, and the results may be sensitive to these choices; a supplementary sensitivity analysis (Section 4.3) tested a range of plausible values for both parameters and found the central Adaptive-vs-Static waste advantage to be directionally robust, though its magnitude is sensitive to the reliability-weight value, and the adaptive-threshold's effect on Adaptive's own waste level did not follow a strictly monotonic pattern at the tested repetition count (n = 20). Second, the charging-station sample used in each simulation run was a single random draw of 1,200 stations from the national dataset; repeating the analysis across multiple independent samples would strengthen confidence that the observed topology effects are not an artifact of one particular sample. Third, the twenty-year observation

window, while empirically grounded in the NEVI/NACS policy timeline, may be too short to observe eventual equilibrium outcomes in slower-diffusing scenarios, particularly within the Linear Grid topology. Finally, both spatial topologies were generated algorithmically rather than fit to formal network models: the Scale-Free Hub topology arises from a preferential-attachment-style heuristic rather than a verified power-law degree distribution, and the Linear Grid topology is more precisely a nearest-neighbor chain than a literal lattice. This stylization is standard in agent-based diffusion modeling, but the topological labels should be read as qualitative descriptors of connectivity structure rather than formal correspondence to canonical network models.

Conclusion

The transition to electrified mobility depends on overcoming substantial indirect network externalities between electric vehicles and public charging infrastructure. This research used an Agent-Based Model to examine how government subsidy structures and physical network topologies jointly influence the resolution of a technological standards competition between NACS and CCS1.

The results suggest that static policy mandates risk generating stranded public assets when markets shift to a new standard. Adaptive policy mitigates this financial waste, but this advantage is highly conditional on the underlying spatial deployment of infrastructure.

The simulation suggests that Scale-Free Hub topologies do not simply accelerate diffusion; instead, they widen the range of possible outcomes, allowing occasional strong shifts toward NACS that a Linear Grid cannot produce. This is consistent with the non-ergodic nature of increasing-returns systems described by Arthur (1989), where small early events can produce large, unpredictable differences in market share. The findings also suggest that accelerating a standard switch is not cost-free: Adaptive policy did not reduce, and may have slightly increased, trip failures among remaining legacy users during the transition.

Taken together, these results suggest that effective infrastructure governance in tipping markets cannot rely on subsidy design alone. Policymakers may benefit from explicitly accounting for the

spatial and topological structure of infrastructure deployment when designing adaptive mechanisms, and from anticipating that even successful policy pivots may impose transitional costs on legacy users that warrant complementary mitigation measures. Given the parameter sensitivities and single-sample design noted above, these findings should be read as suggestive of the mechanisms at play rather than as precise, generalizable forecasts of real-world NACS adoption timelines.

Appendix A

Originally, Hypothesis 1 predicted that cumulative subsidy waste would rise under Static policy without reference to topology; Hypothesis 2 predicted that Adaptive policy would unconditionally reduce both waste and consumer friction; and Hypothesis 3 predicted that Scale-Free Hub topology would accelerate diffusion toward a higher median outcome. Initial simulation runs showed that the Adaptive policy's effect on waste was present only in Scale-Free Hubs (where the 25% market-share threshold was reachable) and absent in Linear Grid (where it was not), that consumer friction did not fall under Adaptive policy, and that topology affected outcome variance rather than the median. The hypotheses presented in the main text were revised accordingly to state these conditions explicitly.

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